

Giacomo Saccaggi

SENIOR AI / ML ENGINEER

Lugano, Switzerland | EU Citizen (Italian) | Valid Swiss Permit



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Senior AI/ML Engineer with 7+ years of experience building scalable, production-ready AI systems bridging software engineering, hardware (IoT), and statistical theory. Proven track record of high-stakes leadership, reducing only this year infrastructure costs by €200,000/year and protecting 25% of corporate digital revenue. Expert in MLOps, Privacy-Preserving Analytics, and Multi-Agent Orchestration.

TECHNICAL SKILLS

- **MLOps & Cloud Infrastructure:** Azure (Certified: AI/ML/Data), AWS (EC2, S3, RDS, SageMaker), GCP, Docker, Kubernetes, Airflow, GitHub Actions (CI/CD).
- **Core Programming:** Python (Expert), SQL (Expert), R (Expert), Kotlin/Java (JetBrains SDK), Scala, VB.net.
- **Data Engineering & Search:** Polars, Spark AQE, Elasticsearch, DuckDB, High-throughput Data Pipelines, JDBC, Presto/Trino, TSQL.
- **Specialized AI:** LLM Orchestration (MCP), Multi-agent Systems, Privacy-Preserving Analytics (HyperLogLog), NLP, Computer Vision.
- **Analytics & Visualization:** PowerBI, Tableau, Kibana, Looker Studio, RAWGraphs, Office Suite (Advanced VBA).
- **Builder Toolkit:** Shell (Bash/Batch), Git, HTML5, CSS3, JavaScript, C/C++ (Arduino/IoT), Django, Flask, Unicorn, Unraid OS.

WORK EXPERIENCE

Nielsen - Technical Owner | Senior AI/ML Engineer

(Lugano, Switzerland | April 2022 – Present) 

- **Cost Optimization & Scalability:** Reduced cloud infrastructure costs by €200,000/year (40-60%) by orchestrating the migration of massive data pipelines to Airflow on Kubernetes and refactoring backends from Pandas to Polars.
- **Strategic Revenue Protection:** Protected products representing 25% of the company's digital revenue following the deprecation of 3rd-party cookies by designing and implementing a deterministic statistical pipeline for audience measurement.
- **High-Throughput Engineering:** Maintained strict daily SLAs processing over 5 billion records per execution by architecting highly performant, distributed computing production systems.
- **Generative AI (AppSense):** Solved complex many-to-many attribution problems for the US Mobile Panel by developing a Siamese Network architecture with BERT encoders and Contrastive Learning, subsequently adapting the architecture for a Bayesian demographic prediction model.
- **Agentic Ecosystem:** Increased data accessibility and operational efficiency by building a unified Model Context Protocol (MCP) gateway architecture and internal AI agents, integrating multi-system enterprise platforms (Airflow, Trino/MDL, AWS, Atlassian, GitLab) for natural language querying.

DXC Technology - Data Scientist & ML Specialist

(Milan, Italy | November 2020 - April 2022) 

- **Enterprise Intelligent Automation:** Delivered a key component of a €7M enterprise automation program by engineering a multi-modal AI platform integrating NLP (Transformers), Computer Vision (OpenCV), and OCR engines (Tesseract) for a major energy provider.
- **Real-Time Predictive Monitoring:** Enabled real-time outlier forecasting in naval fleet predictive maintenance and streaming error forecasting for a global OTT media platform by architecting unsupervised anomaly detection models using GANs (AnoGAN).

TK Soluzioni - Data Scientist

(Saronno, Italy | January 2020 - November 2020) 

- **Statistical Modeling & Optimization:** Optimized system performance metrics for a major publishing corporation by designing and implementing Bayesian statistical models alongside linear and multinomial optimization algorithms.
- **Full-Stack AI Integration:** Powered real-time UI components by seamlessly embedding predictive and optimization models into the client's core monitoring software and exposing API endpoints.

Full Stack & Data Engineer — Independent Freelance

(Milan, Italy | January 2018 - December 2019) 

- Architected and deployed custom web applications, REST APIs, and database schemas for mid-sized clients across diverse industries.
- Engineered automated data ingestion pipelines and optimized query performance using Python, Flask, and SQL.

EDUCATION

- **M.Sc. Statistics** - Bicocca University of Milan (2017/2018 – 2019) | Thesis: Separate collection of waste, a classification problem
- **B.Sc. Quantitative Methods** - Catholic University of Milan (2014/2015 – 2017) | Thesis: Prediction of a football competition

KEY OPEN SOURCE & FULL-STACK PROJECTS

- **scomp-link (PyPI):** Reduced library import time from 5.2s to 5ms (99% optimization) by implementing a custom Lazy Loading system for an end-to-end ML toolkit. | [PyPI](#) | [GitHub](#)
- **Celebrimbot:** Developed an autonomous, multi-agent AI coding assistant for JetBrains IDEs featuring an offline-first architecture with local LLM inference. | [GitHub](#) | [JetBrains](#)
- **CardinalityKit:** Created a comprehensive Python library for Privacy-Preserving Analytics implementing probabilistic sketching algorithms (HyperLogLog, HyperReal). | [GitHub](#)
- **Starfleet Engineering (MCP):** Developed a guide and framework for building custom Model Context Protocol servers. | [Project Guide](#)
- **ProjectCompass:** Automated analysis scheduling and visualization by building a containerized execution engine integrated with Docker and GitHub Container Registry. | [GitHub](#)

EXTRA & CERTIFICATIONS

- **Certifications:** Microsoft Certified: Azure AI Fundamentals, Azure Machine Learning, Computer Vision, NLP. Google Cloud Digital Transformation.
- **Languages:** Italian (Native), English (Proficiency), Spanish (Intermediate), German (Beginner).
- **Interests:** Hobby Farmer at the Family's Agricultural Business; IoT Hobbyist developing custom Smart Devices; Licensed Sailor with skippering/crew experience; WSET Qualifications.